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Ch.9 Human Digestive System

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. Where does chemical digestion of carbohydrates begin?

Answer: (d) Mouth — Salivary amylase begins starch digestion in the mouth.

2. Which salivary enzyme starts starch breakdown?

Answer: (b) Amylase (ptyalin) — Ptyalin hydrolyses starch into smaller sugars.

3. What prevents food entering the trachea during swallowing?

Answer: (a) Epiglottis — The epiglottis covers the glottis during swallowing.

4. Why does stomach enzyme activity drop when pH rises?

Answer: (c) Enzymes need acidic pH to work — Pepsin works best in acidic conditions; higher pH reduces its activity.

5. Which change would most affect protein digestion?

Answer: (c) Inhibiting pepsin production — Pepsin begins protein digestion in the stomach.

6. Why is lipase not active in the stomach?

Answer: (b) It needs alkaline pH to work — Pancreatic lipase works best in the alkaline small intestine; the stomach is acidic.

7. Which stomach secretion activates pepsin and kills bacteria?

Answer: (b) Hydrochloric acid (HCl) — HCl converts pepsinogen to pepsin and provides an acidic antimicrobial environment.

8. Why is segmentation important in the small intestine?

Answer: (c) It mixes food with digestive juices — Segmentation churns chyme and exposes it to enzymes and absorptive surfaces.

9. What is the function of villi and microvilli?

Answer: (b) Increase surface area for absorption — They greatly enlarge the absorptive surface of the small intestine.

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10. Which best explains the liver's role in digestion?

Answer: (c) It produces bile for fat digestion — Bile emulsifies fats, aiding lipase action.

SECTION 2: SHORT QUESTIONS

1. What is the main function of the digestive system?

To ingest food, break it down mechanically and chemically, absorb nutrients, and eliminate undigested waste.

2. What is the mode of action of saliva in the mouth?

Saliva moistens and lubricates food with water and mucus, forms a bolus, and contains salivary amylase (ptyalin) that begins starch digestion. Bicarbonate buffers the oral cavity and thiocyanate ions help kill microorganisms.

3. What is the role of the tongue?

The tongue mixes food with saliva, shapes the bolus, helps swallowing, and contains taste buds.

4. What role does the epiglottis play during swallowing?

It covers the opening of the larynx/trachea, directing food into the oesophagus and preventing aspiration.

5. What is the composition of gastric juice?

Gastric juice is the combined secretion of gastric glands: hydrochloric acid (from parietal/oxyntic cells), pepsinogen (from chief cells, activated to pepsin), mucus, water and salts.

6. Why is HCl important in the stomach?

HCl adjusts stomach pH to about 2–3, activates pepsinogen to pepsin, softens food, kills many microorganisms and provides the acidic optimum for pepsin.

7. Differentiate between bolus and chyme.

Bolus	Chyme
Form: Soft rounded mass of chewed food mixed with saliva.	**Form:** Semi-fluid acidic mixture of partially digested food and gastric juice.
Location: Mouth/oesophagus.	**Location:** Stomach and small intestine.

8. Which organ produces bile and what is its function?

The liver produces bile, stored and concentrated in the gallbladder. Bile salts emulsify fats into small droplets, increasing surface area for lipase; they do not contain digestive enzymes.

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9. Differentiate between physical and chemical digestion.

Physical digestion	Chemical digestion
Process: Mechanical breakdown: chewing, churning, segmentation.	**Process:** Enzymatic hydrolysis of macromolecules.
Result: Smaller pieces, larger surface area.	**Result:** Monomers: sugars, amino acids, fatty acids, glycerol.

10. What is emulsification of fats?

Bile salts break large fat globules into tiny droplets, increasing surface area for lipase. It is a physical process, not enzymatic digestion.

11. What is the role of the pyloric sphincter?

It controls the release of chyme from the stomach into the duodenum, preventing overload and backflow.

12. How do villi and microvilli help absorption?

They increase surface area; villi contain capillaries and lacteals for nutrient uptake, and microvilli form the brush border with enzymes and transporters.

13. What are the main functions of the large intestine?

Absorption of water, salts and vitamin K; formation and temporary storage of faeces; mucus secretion; and housing bacteria (e.g., E. coli) that produce vitamin K and some amino acids.

14. What causes jaundice in the digestive system?

Jaundice is yellowing due to excess bilirubin in blood, often from liver disease, bile-duct obstruction or excessive RBC breakdown.

15. How does stress negatively impact digestion?

Stress can alter motility and secretion, reduce blood flow to the gut, increase acid-related discomfort, and disturb appetite and gut microbiota.

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LONG QUESTIONS

1. Explain complete digestion from ingestion to egestion, including mechanical and chemical digestion.

> Mouth: chewing + salivary amylase → Pharynx/oesophagus: swallowing + peristalsis → Stomach: churning + HCl/pepsin → Small intestine: bile/pancreatic enzymes + brush-border enzymes → Absorption via villi → Large intestine: water/salt absorption → Rectum/anus: egestion

- **Mouth:** teeth mechanically break food; salivary amylase begins starch digestion; food becomes a bolus.

- **Stomach:** churning mixes food with gastric juice; HCl activates pepsinogen to pepsin, which begins protein digestion; food becomes acidic chyme.
- **Small intestine:** bile emulsifies fats; pancreatic amylase, trypsin/chymotrypsin, lipase and nucleases digest carbohydrates, proteins, fats and nucleic acids; intestinal enzymes complete digestion; chyme becomes chyle.
- **Absorption:** villi and microvilli absorb monosaccharides, amino acids, fatty acids, glycerol, vitamins, minerals and water.
- **Large intestine:** absorbs water, salts and vitamin K; forms and stores faeces; egestion removes undigested waste.

2. Describe the structure and function of the stomach in digestion.

The stomach is an elastic, U-shaped muscular bag between the oesophagus and duodenum, with cardiac, fundus and pyloric portions. Its wall has four layers: mucosa (with gastric glands), submucosa, muscularis externa (three muscle layers) and serosa.

- **Mechanical digestion:** vigorous churning mixes food with secretions and melts solid fats.
- **Chemical digestion:** parietal cells secrete HCl (pH about 2–3), which activates pepsinogen and kills microbes; chief cells secrete pepsinogen, converted to pepsin to digest proteins into peptides.
- **Protection:** mucus coats the lining against acid and enzymes.
- **Regulation:** the pyloric sphincter releases chyme gradually into the duodenum.

After 3–4 hours the stomach contents become semi-liquid acidic chyme.

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3. Compare the roles of the small and large intestine in digestion.

Small intestine	Large intestine
Main role: final digestion and absorption of nutrients.	**Main role:** absorption of water, salts and vitamin K; formation and storage of faeces.
Parts: duodenum, jejunum, ileum (about 6 m).	**Parts:** caecum, colon, rectum (about 1 m).
Enzymes: pancreatic and intestinal enzymes complete digestion.	**Enzymes:** no major digestive enzymes; bacterial action produces vitamin K.
Surface: circular folds, villi and microvilli greatly increase absorption.	**Surface:** no villi; goblet cells secrete mucus.
Products: chyle; nutrients enter blood/lymph.	**Products:** faeces; waste eliminated through anus.

4. Explain absorption of food from the small intestine.

Absorption occurs mainly in the jejunum and ileum, whose circular folds, villi and microvilli provide a huge surface area. Each villus contains blood capillaries and a lacteal (lymph vessel).

- **Monosaccharides and amino acids:** absorbed by diffusion/active transport into epithelial cells, then into blood capillaries; the hepatic portal vein carries them to the liver for processing/storage.
- **Fatty acids and glycerol:** absorbed by passive transport; inside epithelial cells they are re-formed into triglycerides and coated with proteins to make chylomicrons.
- **Chylomicrons:** enter lacteals, travel through the thoracic lymphatic duct and enter the bloodstream.
- **Water, vitamins and minerals:** absorbed by osmosis, diffusion or active transport.

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5. Discuss accessory organs and their contributions to digestion.

Organ	Contribution
Liver	Produces bile (emulsifies fats); processes absorbed nutrients; stores glycogen, vitamins and minerals; detoxifies harmful substances; synthesizes plasma proteins.
Gallbladder	Stores and concentrates bile; releases it into the duodenum when fatty food arrives.
Pancreas	Exocrine portion secretes alkaline pancreatic juice containing bicarbonate and enzymes: amylase (carbohydrates), trypsin/chymotrypsin/carboxypeptidase (proteins), lipase (fats), nucleases (nucleic acids). Endocrine portion secretes insulin/glucagon.

Secretin and cholecystokinin (CCK) from the duodenum stimulate pancreatic juice and bile release.

6. Describe hormonal and nervous regulation of gastric acid secretion.

Cephalic phase

The sight, smell, taste or thought of food activates the medulla, which sends vagal nerve impulses to gastric glands, stimulating a small amount of gastric juice.

Gastric phase

Food arrival distends the stomach and proteins stimulate endocrine cells to release gastrin. Gastrin travels in blood to gastric glands, promoting more HCl and pepsin secretion and powerful contractions.

Intestinal phase / inhibition

When chyme moves into the small intestine, somatostatin stops HCl release. This negative feedback prevents excess acid and allows the intestine to process chyme gradually.

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INQUISITIVE QUESTIONS

1. Why does the small intestine need both peristalsis and segmentation?

Segmentation mixes chyme with enzymes and exposes it to the absorptive surface; peristalsis propels contents forward. Together they ensure efficient digestion and movement.

2. How does the liver help digestion without using enzymes?

It produces bile, whose salts emulsify fats into small droplets. Bile salts are not enzymes but greatly increase the surface area on which lipase can act.

3. Why do we need bile if we already have fat-digesting enzymes?

Lipase acts at the surface of fat droplets. Bile emulsifies large fat globules into small droplets, greatly increasing surface area for lipase.

4. How does the pancreas “know” when to release enzymes?

Acidic chyme stimulates secretin release, and fats/amino acids stimulate CCK release. These hormones signal the pancreas to secrete bicarbonate and enzymes; vagal signals also contribute.

5. Why are pancreatic secretions alkaline rather than acidic?

Bicarbonate neutralizes acidic chyme from the stomach, protects the duodenum and provides the alkaline pH needed by pancreatic enzymes.

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